Figure 7.8. This is a degenerate distribution, since all the possible values are assigned a probability of 0 except for one value. The outcome values that are not assigned 0 probability are called the probability distribution's support. The degenerate distribution has a support of 1 element (in this case, the value 0).

Figure 7.9. We've created a function that takes a discrete distribution and updates it based on observed rewards. This function is performing a kind of approximate Bayesian inference by updating a prior distribution into a posterior distribution. Starting from a uniform distribution (on top, we observe some rewards and we get a peaked distribution at 0 (shown in the middle), and then we observe even more rewards (all zeros), and the distribution becomes a narrow, normal-like distribution (as shown on the bottom).

Figure 7.10. This figure shows how a uniform distribution changes with lower or higher values for gamma (the discount factor).

Figure 7.11. The update_dist function redistributes probability from neighbors toward the observed reward value.

Figure 7.12. This is the result of updating an initially uniform probability distribution after observing a single reward. Some probability mass is redistributed toward the support element corresponding to the observed reward.

Figure 7.13. This is the result of updating an initially uniform probability distribution after observing a sequence of different rewards. Each "peak" in the distribution corresponds to an observed reward.

Figure 7.14. The result of updating an initially uniform probability distribution after observing the same reward multiple times. The uniform distribution converges toward a normal-like distribution.

Figure 7.15. The Dist-DQN accepts a 128-element state vector and produces 3 separate 51-element probability distribution vectors, which then get stacked into a single 3 × 51 matrix.

Figure 7.16. A generative model can be a probabilistic model that trains by maximizing the probability that it generates samples that are similar to some empirical data set. Training happens in an iterative loop where the empirical data is supplied to the generative model, which tries to